PHY 504

Assignment #8

due 30 October 2026

1. For the inverse-square potential $V(r) = -k/r^2$ with $k > 0$ (corresponding to an attractive inverse-cube force)

   (a) Sketch the effective potential for $\ell^2 > 2mk$, $\ell^2 = 2mk$, and $\ell^2 < 2mk$, where m is the mass and ℓ is the angular momentum. Then show that the orbits can have the following possible forms:

   $$\frac{1}{r} = A\cos\beta(\theta - \theta_0) \qquad \frac{1}{r} = A(\theta - \theta_0)$$
   $$\frac{1}{r} = A\cosh\beta(\theta - \theta_0) \qquad \frac{1}{r} = \frac{1}{r_o}e^{\pm\beta\theta}$$
   $$\frac{1}{r} = A\sinh\beta(\theta - \theta_0)$$

   (b) For what range of E and ℓ does each of the above types of motion occur? Express the constants A and β in terms of $E, \ell, m,$ and k, where applicable. Explain why r_0 in the exponential solution is not fixed by E and ℓ. Treat the case $\ell^2 = 2mk$, $E = 0$ separately and determine the possible circular orbits. Distinguish scattering, bounded, and plunge orbits.

   (c) Plot a typical orbit of each type using Mathematica (use `PolarPlot`) or Python. Restrict each plot to regions where $r > 0$; don't continue trajectories through $r = 0$. Relate each plot to your effective-potential classification.

2. Consider a modified gravitational potential, with $k > 0$ and $b > 0$,

   $$V(r) = -\frac{k}{r} - \frac{b}{r^2}.$$

   (a) Sketch the effective potential for $b > \ell^2/2m$ and for $b < \ell^2/2m$. Based on your sketches, classify the possible motions as the energy varies. What kinds of orbits do you expect in each case?

   (b) For $b < \ell^2/(2m)$, determine the minimum allowed energy $E_{\min}(\ell)$. Find the shape $r(\theta)$ of a bound orbit of this potential for $E_{\min} \le E < 0$. Check that your result reduces to a Kepler orbit when $b \to 0$.

(c) Plot at least three radial periods of a sample noncircular bound orbit. At what angular speed does the orbit's perihelion precess? You may assume b is very small. Find the exact precession of the perihelion between successive radial minima, relative to 2π, and determine whether the precession is prograde (positive) or retrograde (negative). Obtain the leading approximation for $\epsilon = 2mb/\ell^2 \ll 1$.

3. For the central potential $V(r) = \frac{1}{2}m\omega^2 r^2$:

   (a) Working in polar coordinates, determine the allowed range of E at fixed ℓ. Solve by quadratures for the shape $r(\theta)$ and show that the orbit is an ellipse with center at the origin. What is the eccentricity in terms of E, ℓ, m, and ω? What is the semimajor axis? Check the orbit using the Cartesian oscillator equations.

   (b) Solve for the motion $r(t), \theta(t)$. What is the orbital period? How are the orbital and angular frequencies related?

   (c) Contrast your solution with the elliptical orbits of the Kepler potential $V(r) = -\frac{k}{r}$ with $k > 0$: compare the location of the force center, the dependence of the period on energy, and the number of radial oscillations per complete orbit. Under what condition does a noncircular bound orbit in problem 2 close?